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SUPERSEDING
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FEDERAL SPECIFICATION

COPPER ALLOY INGOTS; BRASS (YELLOW, HIGH STRENGTH) MANGANESE AND MANGANESE-ALUMINUM BRONZE (FOR REMELTING)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for high strength yellow brass (manganese and manganese-aluminum bronze) alloy ingots for remelting and recasting into specified shapes.

1.2 Classification.

1.2.1 Composition. The ingots covered by this specification shall be of the following alloys, as specified (see 6.1) and shall be of the composition listed in table I.

Alloy A
Alloy B
Alloy C
Alloy D
Alloy E
Alloy F

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Locked-Corner.

Federal Standard:

Fed. Test Method Std. No. 151 - Metals, Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specification required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

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FSC 9650

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Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Chemical composition. Ingots shall conform to the chemical requirements shown in table I.

3.1.1 Analysis shall be made regularly only for the elements specifically mentioned in table I. If, however, the presence of other elements is suspected or indicated in the course of routine analysis, further analysis shall be made to determine that the total of other elements is not in excess of the limits specified.

3.1.2 The contractor shall furnish an analysis of each lot showing the percentage of each of the elements specified in table I.

3.1.3 Chemical analysis of the individual lots by the contractor may be waived at the discretion of the procuring agency, provided the manufacturer's method of composition control is acceptable to it or that all the material in the lot can be identified as being from melts previously analyzed and found to be in conformance with chemical compsition requirements of the alloy.

3.2 Mechanical properties.

3.2.1 Ingots. When remelted and cast in sand, ingots shall conform to the properties shown in table II.

3.3 Weights. Unless otherwise specified (see 6.1), ingots shall be furnished in commercial weights.

3.4 Marking for identification. The manufacturer's name or brand and the alloy shall be cast or stamped on each ingot. Each ingot shall bear its proper heat or melt number either stamped on or cast in, so that its identity will be clear at the point of delivery.

3.5 Workmanship. Ingots shall be of uniform quality, clean and free from excessive dirt, dross, slag inclusions, surface exudations, burned or gassed metal and shall not exhibit a "puffed" condition.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For purpose of sampling, a lot shall consist of not more than 100,000 pounds of ingots from a single melt or not more than 2,000 pounds of ingots poured from a continuous melting furnace.

4.2.2 Sampling for inspection. For the purpose of inspection specified in 4.1, the Government inspector shall select representative sample ingots from each lot as shown in table III.

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TABLE I. Chemical Composition

Alloy	Copper	Aluminum	Manganese	Iron	Tin	Lead	(Maximum) Nickel	Zinc
	Percent	Percent	Percent	Percent	Percent	Percent	Percent	Percent
A	55-60	0.5-1.5	1.50 1/	0.4-2.0	1.00 1/, 2/	0.30 1/	0.50	Remainder
B and C	60-68	3.0-7.5	2.5-5.0	2.0-4.0	0.10 1/	0.10 1/	1.0	Remainder
D	56-62	0.25-1.0	0.1-0.5	0.75-1.5	0.5-1.0	0.5-1.0	0.50	Remainder
E	66-68	4.5-6.0	3.0-5.0	2.0-4.0	0.10 1/	0.10 1/	1.0	Remainder
F	56-61	0.7-1.0	1.50 1/	0.4-2.0	1.00 1/, 2/	0.30 1/	0.50	38-40

1/ Maximum
2/ A minimum of 0.50 percent tin shall be specified (see 6.1), when the purpose for which manganese bronze is to be used requires maximum resistance to dezincification.

TABLE II. Mechanical Properties

Alloy	(Minimum) Tensile strength	(Minimum) 1/ Yield strength	Elongation in 2 Inches
	P.s.i.	P.s.i.	P.s.i.
A	65,000		20
B	90,000	45,000	18
C	110,000	60,000	12
D	60,000	20,000	15
E	90,000	45,000	18
F	65,000		20

1/ 0.5 percent extension under load or 2.2 percent offset.

TABLE III. Sampling for Inspection

Number of ingots in lot	Number of ingots in sample
Under 12	All
12 - 100	11
Over 100	12

4.2.3 Sampling for tests. At least four ingots shall be selected by the Government inspector from each lot for tests.

4.2.3.1 Chemical analysis. Samples from two of the ingots selected in accordance with 4.2.3 shall be obtained by drilling entirely through two segments of each ingot (one at the end and one centrally located). Drillings shall include metal at both surfaces. Drillings shall not be mixed, except that drillings taken from one ingot may be thoroughly mixed to make one sample. Each sample shall consist of not less than two ounces.

4.2.3.2 Tension tests.

4.2.3.2.1 After the samples for chemical analysis have been taken, at least two test coupons shall be cast in sand from the ingots selected in accordance with 4.2.3. When practicable, the Government inspector shall witness the pouring of the test bars, and their removal from the mold. He shall take such action as necessary to identify the castings and test bars with the heat or lot number assigned and shall stamp each test bar for future identification with the proper heat or lot number and suitable identification stamp.

4.2.3.2.2 When it is not practicable for the Government inspector to witness the pouring of the test coupons he shall require the manufacturer to establish and maintain procedures and records that will assure the inspector that test coupons are poured, processed, and identified as to the proper heat, melt or lot number of castings they represent.

4.2.3.2.3 When the Government inspector has reasonable doubt as to the identity of the test coupons, drillings for chemical analysis shall be taken from the coupons and the castings purported to be from the same lot for check analysis, to assist the inspector in establishing identity. If the results of any check analysis are conclusive that the test coupons do not represent the lot involved, the Government inspector shall take proper steps to have the manufacturer correct his procedures and maintain proper identity of test bars.

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4.2.3.2.4 Unless otherwise specified in the contract or order, the chilling of test coupons shall be cause of rejection of the material represented.

4.2.3.2.5 The test coupon shall conform to the form and dimensions shown on figure 1. Other test bar designs may be used if approved by the bureau or agency concerned. The specimen machined therefrom shall conform to the form and dimensions shown in Federal Test Method Std. No. 151.

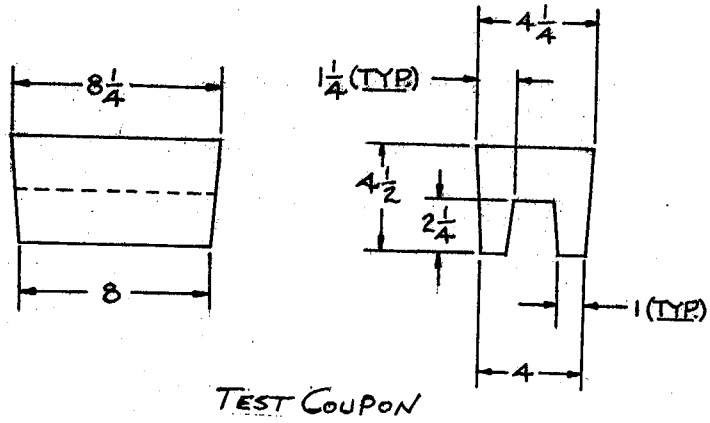


FIGURE 1

4.3 Inspection. Each ingot selected in accordance with 4.2.2 shall be inspected by the Government inspector to determine conformance with this specification. If one or more sample ingots fail to conform to this specification, the entire lot shall be rejected. A rejected lot may be resubmitted, after the manufacturer has reinspected each ingot in the lot for the deficiency noted, and has removed all nonconforming ingots.

4.4 Test procedures.

4.4.1 Chemical analysis. Each of the samples obtained in accordance with 4.2.3.1 shall be forwarded to a Government laboratory designated by the procuring activity of a laboratory satisfactory to the Government inspector for chemical analysis to determine conformance with 3.1.1. If either of the samples fail to conform to this specification, the lot shall be rejected.

4.4.2 Tension tests. One of the test specimens obtained in accordance with 4.2.3.2 shall be pulled in tension in accordance with Federal Test Method Std. No. 151 to determine conformation with 3.2. If the specimen fails to conform to this specification, except as provided in 4.4.3.1, the lot shall be rejected.

4.4.2.1 If any test specimen contains flaws which result in the failure of the specimen, an additional specimen from the same melt may be selected to replace it at the discretion of the Government inspector. The nature of the flaw or defect in the discarded specimen, when deemed characteristic of the material under inspection shall be taken into consideration when determining the acceptability of the material.

4.4.3 Retests. Retests for mechanical properties will be permitted in accordance with Federal Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be Level A or C as specified (see 6.1(e)).

5.1.1 Level A. Ingots shall be packed in a snug fitting cleated-plywood box, overseas type, style A or B conforming to PPP-B-601; or in a snug fitting nailed wood box, class 2, style 2, 2-1/2, 3 or 7 conforming to PPP-B-621. Weight of contents shall not exceed 1,000 pounds. Banding shall be accomplished in accordance with the appendix of the applicable specification.

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5.1.1.1 Containers shall only be packed with ingots of like alloys and Federal serial numbers.

5.1.2 Level C. Ingots shall be palletized in accordance with the requirements of MIL-STD-147. Gross load shall not exceed 3,500 pounds.

5.1.2.1 Palletization shall include only ingots of like alloys and Federal serial numbers.

5.1.2.2 Less than adequate pallet loads shall be banded to snug fitting skids of sufficient strength to meet handling, storage and shipment requirements.

5.2 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Alloy required (see 1.2).
(c) Weights of the ingots, if not to be furnished in commercial weights (see 3.3).
(d) When a minimum of 0.50 percent tin is required (see table I).
(e) Whether packing shall be Level A or Level C (see 5.1.1 and 5.1.2).
(f) Marking required (see 5.2).

6.2 Quality control. The requirements specified in 4.2.1 for a lot to consist of a single melt may be modified upon application by the contractor or prospective contractor to the procuring activity, via the Government inspector, giving detailed information that the contractor or prospective contractor has demonstrated his ability to produce ingots of uniform quality in accordance with this specification.

6.3 The packing and marking specified herein apply to direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

6.4 Federal specification do not include all types, classes, grades, and sizes, of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, classes, grades, and sizes which are suitable for Federal Government requirements.

6.5 Ingots covered by this specification are equivalent to castings of the following composition:

Alloy	Federal Specification	Federal Specification	Copper Development Association Equivalent Number
A		QQ-B-726 (Class-A)	#865
B	MIL-B-16522 (Class-2)	QQ-B-726 (Class-B)	#862
C	MIL-B-16522 (Class-1)	QQ-B-726 (Class-C)	#863
D (Leaded)		QQ-B-726 (Class-D)	#864
E	MIL-B-16522 (Class-3)		#861
F (Modified Alloy-A)			#865

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